MACHINE LEARNING

LESSON 01:

Introduction to Machine Learning

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USEFUL MATERIALS

- This class is inspired by and is adapted from the Coursera Machine Learning course by Andrew Ng (coursera.org/learn/machine-learning)

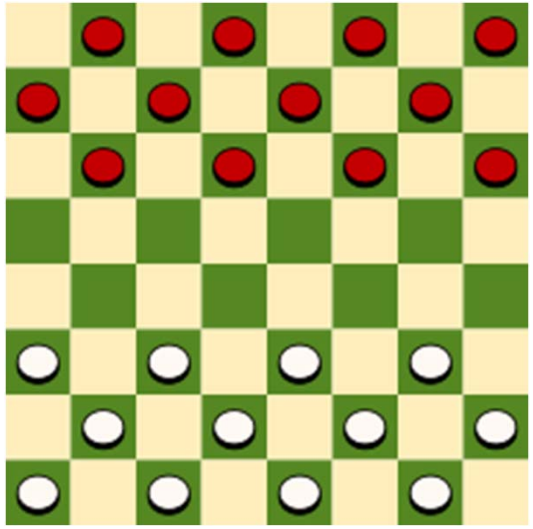
- Python Environment:
  - Python environment using Anaconda (anaconda.com)
  - Pycharm IDE (jetbrains.com/pycharm/download)
  - Google Colab

- Machine Learning Libraries/Frameworks:
  - Numpy (numpy.org)
  - Scikit-learn (scikit-learn.org)
  - Pytorch (pytorch.org)

- Python and Numpy Review (from Standford CS231n by Justin Johnson):
  - cs231n.github.io/python-numpy-tutorial

WHAT IS MACHINE LEARNING?

- **Arthur Samuel (1959)** Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.



WHAT IS MACHINE LEARNING?

- **Tom Mitchell (1998)** Well-posed Learning Problem: A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.

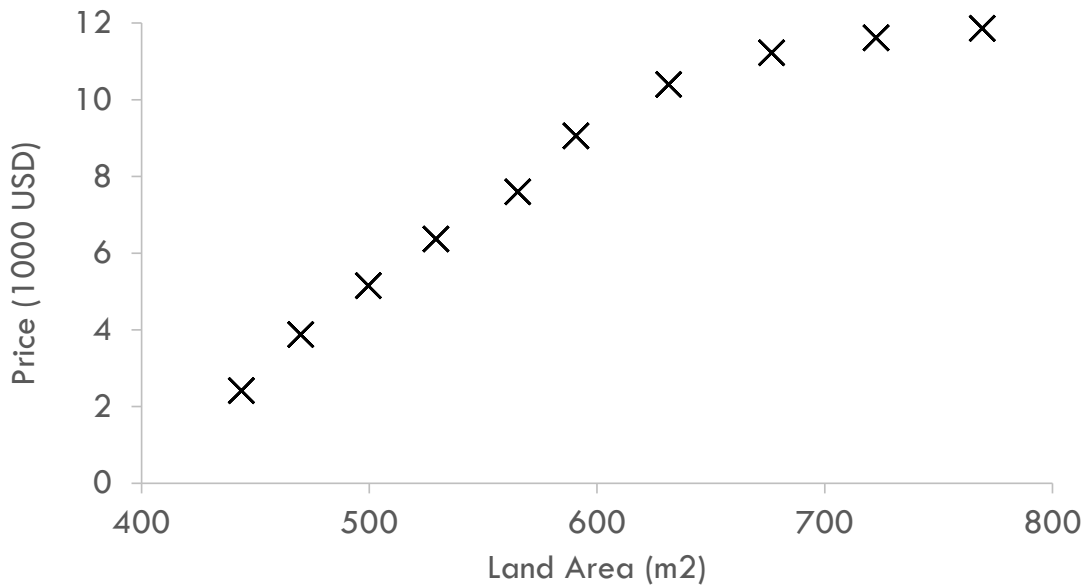
1. Classifying emails as spam or not spam
2. Watching you label emails as spam or not spam
3. The number (or fraction) of emails correctly classified as spam/not spam

MACHINE LEARNING ALGORITHMS

- Supervised Learning
- Unsupervised Learning

SUPERVISED LEARNING

- Example 1: Land price prediction

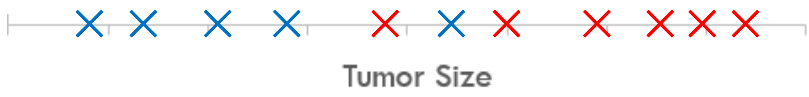


Regression:
predict continuous valued output

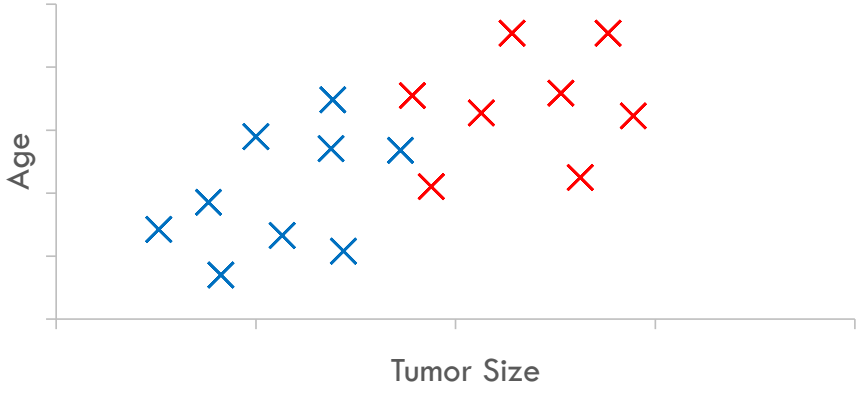
Supervised Learning ⇒ the right answers are given

SUPERVISED LEARNING

- Example 2: Cancer detection
  - 1 variable: tumor size



  - 2 variables: tumor size and patient's age



Classification: predict discrete valued output

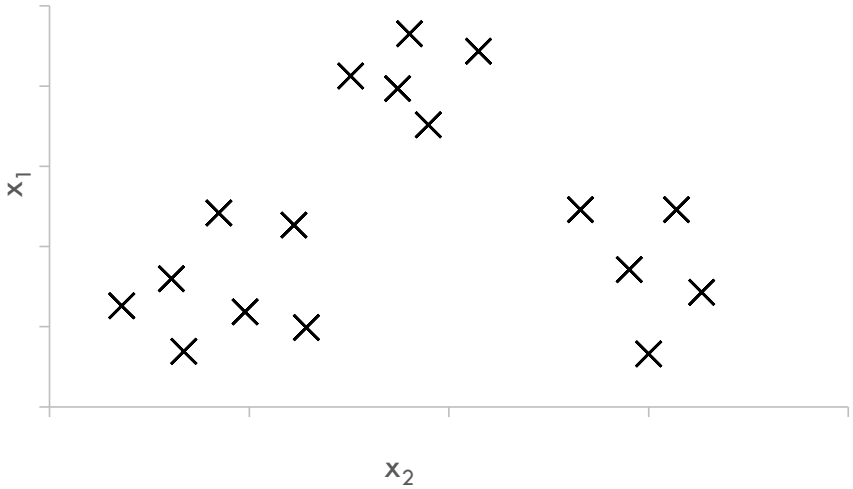
SUPERVISED LEARNING

- You're running a company, and you want to develop learning algorithms to address each of two problems.
  - Problem 1: You have a large inventory of identical items. You want to predict how many of these items will sell over the next 3 months.
  - Problem 2: You'd like software to examine individual customer accounts, and for each account decide if it has been hacked/compromised.

Should you treat these as classification or as regression problems?

UNSUPERVISED LEARNING

- Example:



Clustering: find a structure or pattern in a collection of uncategorized data

Unsupervised Learning ⇒ no label is given

UNSUPERVISED LEARNING

- More examples:
  - item recommendation/suggestion
  - anomaly detection
  - dimensionality reduction

SUPERVISED VS UNSUPERVISED

- Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not
- Given a set of news articles found on the web, group them into set of articles about the same story
- Given a database of customer data, automatically discover market segments and group customers into different market segments
- Given a dataset of images of 3 species of birds, learn to identify which specie an image of a bird belong to